Lagrange's Theorem

office hours:

Find algebraic constraints on the size of subgroups

Let G be a group & H a subgroup.

Def a left coset of H with representative g is $gH = \{gh : h \in H\}$

Is that a subgroup? usually no.

If $g \in H$ then $gH = H$

right coset $Hg = \{hg : h \in H\}$

Examples

① $\mathbb{Z}_6 = (\{0,1,2,3,4,5\}, +)$

$H = \{0,3\}$

$0+H = 3+H = \{0,3\}$

$1+H = 4+H = \{1,4\}$

$2+H = 5+H = \{2,5\}$

② D_6

$\langle r, s \rangle$ with $r^6 = e$, $s^2 = e$, $srs = r^{-1}$

$D_6 = \{e, r, r^2, r^3, r^4, r^5, s, sr, sr^2, sr^3, sr^4, sr^5\}$

$H = \{e, r^2, r^4\}$, $sH = \{s, sr^2, sr^4\}$, $rH = \{r, r^3, r^5\}$

sr^2H "

$srH = \{sr, sr^3, sr^5\}$

three important observations

1) Union of cosets is whole group

2) intersection of cosets is $\emptyset$ or entire set

3) All the cosets have the same size

1) <u>Claim</u> Every element of G lies in a coset of H.

<u>Pf</u> Let $g \in G$ $\quad g \in gH$ $\square$ ✓

3) Claim there is a bijection between the elements of H & gH.

Pf Let $f : H \to gH$ where $f(x) = gx$

onto By the definition of right coset this is a surjection.

1 to 1 Suppose $f(x) = f(y)$ want to show $x = y$

$\Rightarrow gx = gy$

$\Rightarrow x = y$ (mult. by g^{-1})

Cor $|H| = |gH|$ for all $g \in G$.

Lemma $g_1 H = g_2 H \iff g_1 \in g_2 H$ $(g_2 \in g_1 H)$

if $g_1 \in g_2 H$: Let $x \in g_1 H$

So $x = g_1 h_1$ for some $h_1 \in H$ but $g_1 = g_2 h_2$ for $h_2 \in H$

So $x = g_2 \underbrace{h_2 h_1}_{\in H} \Rightarrow x \in g_2 H$

So any element of $g_1 H$ is an element of $g_2 H$

$\Rightarrow g_1 H \subseteq g_2 H$ but $|g_1 H| = |g_2 H|$

So $g_1 H = g_2 H$.

if $g_1 H = g_2 H$ want to show is $g_1 \in g_2 H$

$g_1 \cdot e \in g_1 H = g_2 H \Rightarrow g_1 e = g_1 \in g_2 H$.

$\square$

(2) Claim $g_1 H = g_2 H \iff g_1 H \cap g_2 H \neq \emptyset$

if $g_1 H = g_2 H$ $\quad g_1 H \cap g_2 H = g_1 H = g_2 H \neq \emptyset$

if $g_1 H \cap g_2 H \neq \emptyset$ want to show $g_1 H = g_2 H$

Let $x \in g_1 H \cap g_2 H$ So $g_1 h_1 = x = g_2 h_2$

where $h_1, h_2 \in H$. So $g_1 = g_2 \underbrace{h_2 h_1^{-1}}_{H}$ (mult by h_1^{-1} on right)

$\Rightarrow g_1 \in g_2 H$ so by lemma
$g_1 H = g_2 H$.

$\square$

Lagrange's Theorem

Let G be a finite group & H a subgroup of G. Let $[G:H]$ be the index of H in G. Then $|G| = [G:H]\,|H|$

Def The index of H in G, $[G:H]$ is the number of ~~right~~ left cosets of H.

Is $gH = Hg$? Not generally

<u>Pf</u> The cosets of H in G are all the same size $|H|$.

The cosets of H in G form a disjoint union of G.

$$|G| = \underbrace{|H| + |H| + |H| + \cdots |H|}_{[G:H] \text{ times}}$$

$$|G| = |H| \, [G:H] \qquad \square$$

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